

GEN211 Probability and Statistics

Lecture 1: Descriptive Statistics I: Data Organization and Data Presentation





Lecture contents

- Introduction
- Data organization
- Data presentation



Introduction



Basic definitions

Statistics is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.

A **variable** is a characteristic or attribute that can assume different values.

Data are the values (measurements or observations) that the variables can assume. Variables whose values are determined by chance are called **random variables**.

Suppose that an insurance company studies its records over the past several years and determines that, on average, 3 out of every 100 automobiles the company insured were involved in accidents during a 1-year period. Although there is no way to predict the specific automobiles that will be involved in an accident (random occurrence), the company can adjust its rates accordingly, since the company knows the general pattern over the long run. (That is, on average, 3% of the insured automobiles will be involved in an accident each year.)

A collection of data values forms a **data set**. Each value in the data set is called a **data value** or a **datum**.



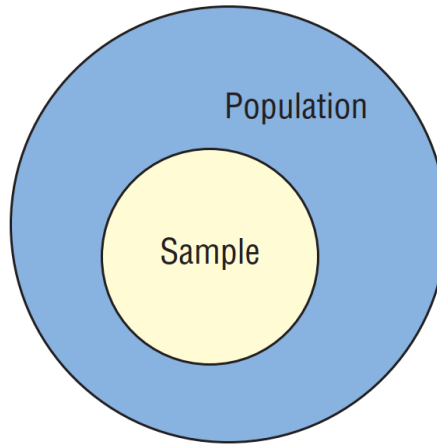
Basic definitions

In statistics it is important to distinguish between a sample and a population.

A **population** consists of all subjects (human or otherwise) that are being studied.

A **sample** is a group of subjects selected from a population.

Population and Sample



Basic definitions

The body of knowledge called statistics is sometimes divided into two main areas, depending on how data are used. The two areas are

1. Descriptive statistics
2. Inferential statistics

Descriptive statistics consists of the collection, organization, summarization, and presentation of data.

In *descriptive statistics* the statistician tries to describe a situation. Consider the national census conducted by the U.S. government every 10 years. Results of this census give you the average age, income, and other characteristics of the U.S. population. To obtain this information, the Census Bureau must have some means to collect relevant data. Once data are collected, the bureau must organize and summarize them. Finally, the bureau needs a means of presenting the data in some meaningful form, such as charts, graphs, or tables.



Basic definitions

In other words, **descriptive statistics** are methods used to summarize, describe, and analyze the main features of a data set. They help provide a clear and general brief overview of the data, making it easier to understand and interpret.

The second area of statistics is called *inferential statistics*.

Inferential statistics consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.

Here, the statistician tries to make inferences from *samples* to *populations*. Inferential statistics uses **probability**, i.e., the chance of an event occurring.



Basic definitions

In other words, **inferential statistics** include the branch of statistics that focuses on making conclusions, predictions, or generalizations about a population based on data collected from a sample. Instead of just describing the data you have, which is addressed in descriptive statistics, inferential statistics helps you infer insights about a larger group that you haven't fully measured.

The science of inferential statistics typically involves:

1. Estimating population parameters

1. Example: Using a sample mean to estimate the true population mean.
2. Tools: confidence intervals.

2. Hypothesis testing

1. Example: Testing whether a new teaching method improves student scores compared to the traditional method.
2. Tools: t-tests, chi-square tests, ANOVA, etc.

3. Making predictions

1. Example: Predicting future sales based on sample data trends.
2. Tools: regression analysis, correlation analysis.

👉 In short:

“Descriptive statistics”: To summarize what you see.

“Inferential statistics”: To make predictions or conclusions beyond what you see.



Example 1

Determine whether descriptive or inferential statistics were used.

- a.* The average jackpot for the top five lottery winners was \$367.6 million.
- b.* A study done by the American Academy of Neurology suggests that older people who had a high caloric diet more than doubled their risk of memory loss.
- c.* Based on a survey of 9317 consumers done by the National Retail Federation, the average amount that consumers spent on Valentine's Day in 2011 was \$116.
- d.* Scientists at the University of Oxford in England found that a good laugh significantly raises a person's pain level tolerance.



Example 1

SOLUTION

- a.* Descriptive statistics were used because this is an average, and it is based on data obtained from the top five lottery winners at this time.
- b.* Inferential statistics were used since this is a generalization made from a sample to a population.
- c.* Descriptive statistics were used since this is an average based on a sample of 9317 respondents.
- d.* Inferential statistics were used since an inference is made from a sample to a population.



Variables and data types

Variables can be classified as qualitative or quantitative.

Qualitative variables are variables that have distinct categories according to some characteristic or attribute.

For example, if subjects are classified according to gender (male or female), then the variable *gender* is qualitative. Other examples of qualitative variables are religious preference and geographic locations.

Quantitative variables are variables that can be counted or measured.

For example, the variable *age* is numerical, and people can be ranked in order according to the value of their ages. Other examples of quantitative variables are heights, weights, and body temperatures.



Variables and data types

1- A qualitative variable (also called a **categorical variable**) is a type of variable that describes **qualities, categories, or attributes**, rather than numerical values.

It represents **labels or names** that classify things into groups. The values cannot be meaningfully added, subtracted, or averaged.

Examples of qualitative variables:

Gender: male, female.

Marital status: single, married, divorced, widowed.

Eye color: blue, brown, green.

Type of car: sedan, SUV, truck.

Nationality: Egyptian, French, Japanese.



Variables and data types

2- A quantitative variable is a variable that represents **numerical values** and can be measured or counted. The values are numbers that express *how much* or *how many* of something.

Unlike **qualitative variables** (categories), **quantitative variables** deal with numbers that you can calculate with (add, subtract, average, etc.).

- **Examples of quantitative variables:**

Age: 21 years, 35 years

Height: 175 cm, 162 cm

Weight: 70 kg, 85 kg

Number of children: 0, 1, 2, 3

Income: \$1000, \$2500



Variables and data types

Quantitative variables can be further classified into two groups: discrete and continuous. *Discrete variables* can be assigned values such as 0, 1, 2, 3 and are said to be *countable*. Examples of discrete variables are the number of children in a family, the number of students in a classroom, and the number of calls received by a switchboard operator each day for a month.

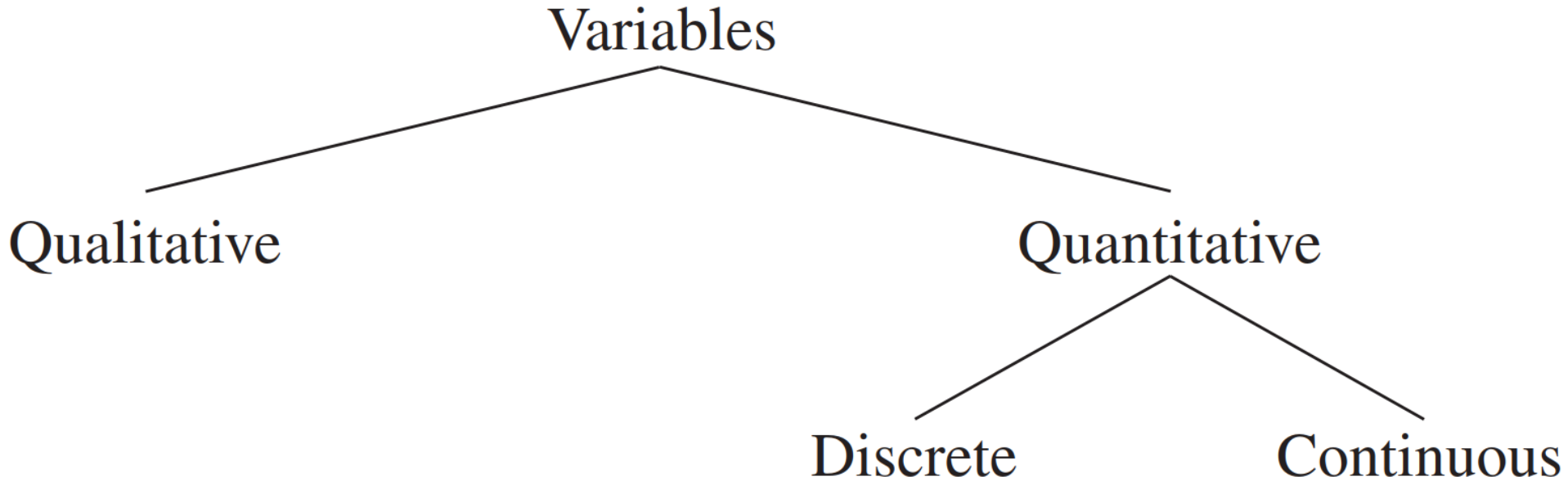
Discrete variables assume values that can be counted.

Continuous variables, by comparison, can assume an infinite number of values in an interval between any two specific values. Temperature, for example, is a continuous variable, since the variable can assume an infinite number of values between any two given temperatures.

Continuous variables can assume an infinite number of values between any two specific values. They are obtained by measuring. They often include fractions and decimals.



Variables and data types



Example 2

Classify each variable as a discrete variable or a continuous variable.

- a.* The highest wind speed of a hurricane
- b.* The weight of baggage on an airplane
- c.* The number of pages in a statistics book
- d.* The amount of money a person spends per year for online purchases

SOLUTION

- a.* Continuous, since wind speed must be measured
- b.* Continuous, since weight is measured
- c.* Discrete, since the number of pages is countable
- d.* Discrete, since the smallest value that money can assume is in cents



Measurement levels

In addition to being classified as qualitative or quantitative, variables can be classified by how they are categorized, counted, or measured. For example, can the data be organized into specific categories, such as area of residence (rural, suburban, or urban)? Can the data values be ranked, such as first place, second place, etc.? Or are the values obtained from measurement, such as heights, IQs, or temperature? This type of classification—i.e., how variables are categorized, counted, or measured—uses **measurement scales**, and four common types of scales are used: nominal, ordinal, interval, and ratio.



Measurement levels

The **nominal level of measurement** classifies data into mutually exclusive (nonoverlapping) categories in which no order or ranking can be imposed on the data.

The first level of measurement is called the *nominal level* of measurement. A sample of college instructors classified according to subject taught (e.g., English, history, psychology, or mathematics) is an example of nominal-level measurement. Classifying survey subjects as male or female is another example of nominal-level measurement. No ranking or order can be placed on the data. Classifying residents according to zip codes is also an example of the nominal level of measurement. Even though numbers are assigned as zip codes, there is no meaningful order or ranking.



Measurement levels

The **ordinal level of measurement** classifies data into categories that can be ranked; however, precise differences between the ranks do not exist.

The next level of measurement is called the *ordinal level*. Data measured at this level can be placed into categories, and these categories can be ordered, or ranked. For example, from student evaluations, guest speakers might be ranked as superior, average, or poor. Floats in a homecoming parade might be ranked as first place, second place, etc. *Note that precise measurement of differences in the ordinal level of measurement does not exist.* For instance, when people are classified according to their build (small, medium, or large), a large variation exists among the individuals in each class.

Other examples of ordinal data are letter grades (A, B, C, D, F).



Measurement levels

The **interval level of measurement** ranks data, and precise differences between units of measure do exist; however, there is no meaningful zero.

The third level of measurement is called the *interval level*. This level differs from the ordinal level in that precise differences do exist between units. For example, many standardized psychological tests yield values measured on an interval scale. IQ is an example of such a variable. There is a meaningful difference of 1 point between an IQ of 109 and an IQ of 110. Temperature is another example of interval measurement, since there is a meaningful difference of 1°F between each unit, such as 72 and 73°F. *One property is lacking in the interval scale: There is no true zero.* For example, IQ tests do not measure people who have no intelligence. For temperature, 0°F does not mean no heat at all.



Measurement levels

The **ratio level of measurement** possesses all the characteristics of interval measurement, and there exists a true zero. In addition, true ratios exist when the same variable is measured on two different members of the population.

The final level of measurement is called the *ratio level*. Examples of ratio scales are those used to measure height, weight, area, and number of phone calls received. Ratio scales have differences between units (1 inch, 1 pound, etc.) and a true zero. In addition, the ratio scale contains a true ratio between values. For example, if one person can lift 200 pounds and another can lift 100 pounds, then the ratio between them is 2 to 1. Put another way, the first person can lift twice as much as the second person.



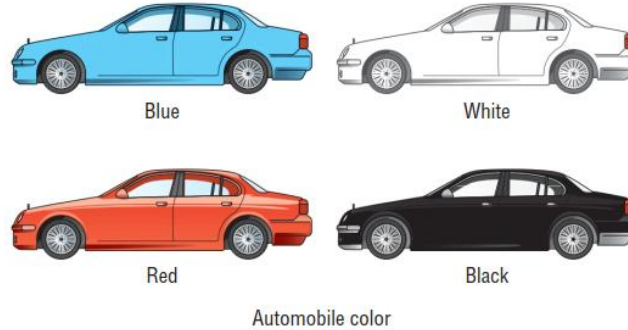
Measurement levels

Examples of Measurement Scales			
Nominal-level data	Ordinal-level data	Interval-level data	Ratio-level data
Zip code Gender (male, female) Eye color (blue, brown, green, hazel) Political affiliation Religious affiliation Major field (mathematics, computers, etc.) Nationality	Grade (A, B, C, D, F) Judging (first place, second place, etc.) Rating scale (poor, good, excellent) Ranking of tennis players	SAT score IQ Temperature	Height Weight Time Salary Age

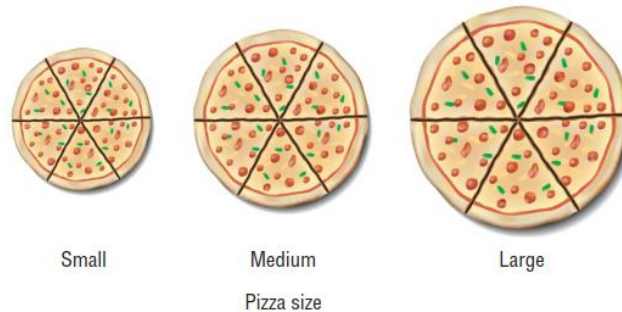


Measurement levels

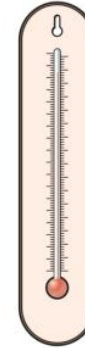
1. Nominal Level



2. Ordinal Level

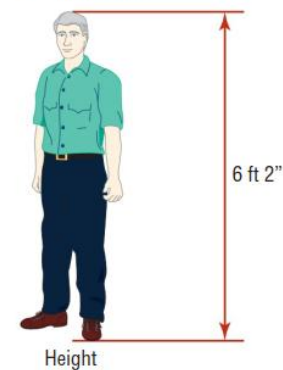


3. Interval Level



Temperature

4. Ratio Level



Example 3

What level of measurement would be used to measure each variable?

- a.* The ages of patients in a local hospital
- b.* The ratings of movies released this month
- c.* Colors of athletic shirts sold by Oak Park Health Club
- d.* Temperatures of hot tubs in local health clubs

SOLUTION

- a.* Ratio
- b.* Ordinal
- c.* Nominal
- d.* Interval



Data Organization



Motivation

When conducting a statistical study, the researcher must gather data for the particular variable under study. For example, if a researcher wishes to study the number of people who were bitten by poisonous snakes in a specific geographic area over the past several years, he or she has to gather the data from various doctors, hospitals, or health departments.

To describe situations, draw conclusions, or make inferences about events, the researcher must organize the data in some meaningful way. The most convenient method of organizing data is to construct a *frequency distribution*.

After organizing the data, the researcher must present them so they can be understood by those who will benefit from reading the study. The most useful method of presenting the data is by constructing *statistical charts* and *graphs*. There are many different types of charts and graphs, and each one has a specific purpose.



Motivation

Data organization is the process of arranging collected data in a systematic, logical, and structured way so that it can be easily understood, analyzed, and interpreted.

When **raw data** is first collected, it's usually **messy and unstructured**. **Organizing** it makes patterns clearer and prepares it for statistical analysis.

By data organization, raw, messy data are turned into structured information (tables, charts, graphs) so it's easier to analyze.



Data organization using a frequency distribution

Suppose a researcher wished to do a study on the ages of the 50 wealthiest people in the world. The researcher first would have to get the data on the ages of the people. In this case, these ages are listed in *Forbes Magazine*. When the data are in original form, they are called **raw data** and are listed next.

45	46	64	57	85
92	51	71	54	48
27	66	76	55	69
54	44	54	75	46
61	68	78	61	83
88	45	89	67	56
81	58	55	62	38
55	56	64	81	38
49	68	91	56	68
46	47	83	71	62



Data organization using a frequency distribution

Since little information can be obtained from looking at raw data, the researcher organizes the data into what is called a *frequency distribution*.

A **frequency distribution** is the organization of raw data in table form, using classes and frequencies.

Each raw data value is placed into a quantitative or qualitative category called a **class**. The **frequency** of a class then is the number of data values contained in a specific class. A frequency distribution is shown for the preceding data set.



Data organization using a frequency distribution

Class limits	Tally	Frequency
27–35	/	1
36–44	///	3
45–53	//// ////	9
54–62	//// //// ////	15
63–71	//// ////	10
72–80	///	3
81–89	//// //	7
90–98	//	2
		<u>50</u>



Data organization using a frequency distribution

Now some general observations can be made from looking at the frequency distribution. For example, it can be stated that the majority of the wealthy people in the study are over 45 years old.

The classes in this distribution are 27–35, 36–44, etc. These values are called *class limits*. The data values 27, 28, 29, 30, 31, 32, 33, 34, 35 can be tallied in the first class; 36, 37, 38, 39, 40, 41, 42, 43, 44 in the second class; and so on.

Two types of frequency distributions that are most often used are the *categorical frequency distribution* and the *grouped frequency distribution*.



Categorical vs. grouped frequency distributions

Categorical Frequency Distributions

The **categorical frequency distribution** is used for data that can be placed in specific categories, such as nominal- or ordinal-level data. For example, data such as political affiliation, religious affiliation, or major field of study would use categorical frequency distributions.

Grouped Frequency Distributions

When the range of the data is large, the data must be grouped into classes that are more than one unit in width, in what is called a **grouped frequency distribution**.



Categorical vs. grouped frequency distributions

A **categorical frequency distribution** is a way of organizing qualitative (categorical) data into a table that shows the number of times (frequency) each category occurs.

It is used when data are labels, names, or categories, not numbers.

Ex: Favorite Fruit of 10 Students : **Raw data (responses)**

Apple, Banana, Orange, Apple, Mango, Banana, Apple, Orange, Apple, Mango

Fruit	Frequency	Relative Frequency
Apple	4	40%
Banana	2	20%
Orange	2	20%
Mango	2	20%



Categorical vs. grouped frequency distributions

A **grouped frequency distribution** is a way of organizing quantitative data (numbers) by dividing the data into intervals (called classes or groups) and showing how many data values fall into each interval.

It is especially useful when the data set is large or when the values vary widely, so listing each individual value is not practical.

Steps to set up a grouped frequency distribution:

- **Find the range** (largest value – smallest value).
- **Choose the number of classes** (intervals) — usually 5 to 20.
- **Calculate class width** (range $\div$ number of classes, then round up).
- **Construct class intervals** (non-overlapping, equal width).
- **Count frequencies** (tally how many values fall into each class).



Categorical vs. grouped frequency distributions

A **categorical frequency distribution**: A table that shows how often each category occurs in the data.

A **grouped frequency distribution**: Organizes numerical data into equal-width intervals (classes) and shows how many values fall into each.



Example 4

Distribution of Blood Types

Twenty-five army inductees were given a blood test to determine their blood type. The data set is

A	B	B	AB	O
O	O	B	AB	B
B	B	O	A	O
A	O	O	O	AB
AB	A	O	B	A

Construct a frequency distribution for the data.



Example 4

SOLUTION

Since the data are categorical, discrete classes can be used. There are four blood types: A, B, O, and AB. These types will be used as the classes for the distribution.

The procedure for constructing a frequency distribution for categorical data is given next.

Step 1 Make a table as shown.

A Class	B Tally	C Frequency	D Percent
A B O AB			



Example 4

SOLUTION

- Step 2** Tally the data and place the results in column B.
- Step 3** Count the tallies and place the results in column C.
- Step 4** Find the percentage of values in each class by using the formula

$$\% = \frac{f}{n} \cdot 100$$

where f = frequency of the class and n = total number of values. For example, in the class of type A blood, the percentage is

$$\% = \frac{5}{25} \cdot 100 = 20\%$$

the decimal equivalent of a percent is called a *relative frequency*.



Example 4

SOLUTION

Step 5 Find the totals for columns C (frequency) and D (percent). The completed table is shown. It is a good idea to add the percent column to make sure it sums to 100%. This column won't always sum to 100% because of rounding.

A Class	B Tally	C Frequency	D Percent
A		5	20
B	//	7	28
O	//	9	36
AB		4	16
		<u>25</u>	<u>100%</u>

For the sample, more people have type O blood than any other type.



Grouped frequency distribution

Procedure Table

Constructing a Grouped Frequency Distribution

- Step 1** Determine the classes.
- Find the highest and lowest values.
 - Find the range.
 - Select the number of classes desired.
 - Find the width by dividing the range by the number of classes and rounding up.
 - Select a starting point (usually the lowest value or any convenient number less than the lowest value); add the width to get the lower limits.
 - Find the upper class limits.
 - Find the boundaries.
- Step 2** Tally the data.
- Step 3** Find the numerical frequencies from the tallies, and find the cumulative frequencies.



Example 5

Record High Temperatures

These data represent the record high temperatures in degrees Fahrenheit ($^{\circ}\text{F}$) for each of the 50 states. Construct a grouped frequency distribution for the data, using 7 classes.

112	100	127	120	134	118	105	110	109	112
110	118	117	116	118	122	114	114	105	109
107	112	114	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	113	120	117	105	110	118	112	114	114



Example 5

SOLUTION

The procedure for constructing a grouped frequency distribution for numerical data follows.

Step 1 Determine the classes.

Find the highest value and lowest value: $H = 134$ and $L = 100$.

Find the range: $R = \text{highest value} - \text{lowest value} = H - L$, so

$$R = 134 - 100 = 34$$

Select the number of classes desired (usually between 5 and 20). In this case, 7 is arbitrarily chosen.

Find the class width by dividing the range by the number of classes.

$$\text{Width} = \frac{R}{\text{number of classes}} = \frac{34}{7} = 4.9$$



Example 5

SOLUTION

Round the answer up to the nearest whole number if there is a remainder:

$$4.9 \approx 5.$$

Select a starting point for the lowest class limit. This can be the smallest data value or any convenient number less than the smallest data value. In this case, 100 is used. Add the width to the lowest score taken as the starting point to get the lower limit of the next class. Keep adding until there are 7 classes, as shown, 100, 105, 110, etc.

Subtract one unit from the lower limit of the second class to get the upper limit of the first class. Then add the width to each upper limit to get all the upper limits.

$$105 - 1 = 104$$

The first class is 100–104, the second class is 105–109, etc.



Example 5

SOLUTION

Find the class boundaries by subtracting 0.5 from each lower class limit and adding 0.5 to each upper class limit:

99.5–104.5, 104.5–109.5, etc.

Step 2 Tally the data.

Step 3 Find the numerical frequencies from the tallies.

The completed frequency distribution is

Class limits	Class boundaries	Tally	Frequency
100–104	99.5–104.5	//	2
105–109	104.5–109.5	////	8
110–114	109.5–114.5	////	18
115–119	114.5–119.5	////	13
120–124	119.5–124.5	////	7
125–129	124.5–129.5	/	1
130–134	129.5–134.5	/	1
			Total 50

The frequency distribution shows that the class 109.5–114.5 contains the largest number of temperatures (18) followed by the class 114.5–119.5 with 13 temperatures. Hence, most of the temperatures (31) fall between 110 and 119°F.



Cumulative frequency distribution

Sometimes it is necessary to use a *cumulative frequency distribution*. A **cumulative frequency distribution** is a distribution that shows the number of data values less than or equal to a specific value (usually an upper boundary). The values are found by adding the frequencies of the classes less than or equal to the upper class boundary of a specific class. This gives an ascending cumulative frequency. In this example, the cumulative frequency for the first class is $0 + 2 = 2$; for the second class it is $0 + 2 + 8 = 10$; for the third class it is $0 + 2 + 8 + 18 = 28$. Naturally, a shorter way to do this would be to just add the cumulative frequency of the class below to the frequency of the given class.



Cumulative frequency distribution

For example,
the cumulative frequency for the number of data values less than 114.5 can be found by adding $10 + 18 = 28$. The cumulative frequency distribution for the data in this example is as follows:

Less than 99.5	0
Less than 104.5	2
Less than 109.5	10
Less than 114.5	28
Less than 119.5	41
Less than 124.5	48
Less than 129.5	49
Less than 134.5	50



Reasons for frequency distribution

The reasons for constructing a frequency distribution are as follows:

1. To organize the data in a meaningful, intelligible way.
2. To enable the reader to determine the nature or shape of the distribution.
3. To facilitate computational procedures for measures of average and spread.
4. To enable the researcher to draw charts and graphs for the presentation of data.
5. To enable the reader to make comparisons among different data sets.



Data Presentation



Presenting data

After you have organized the data into a frequency distribution, you can present them in graphical form. The purpose of graphs in statistics is to convey the data to the viewers in pictorial form. It is easier for most people to comprehend the meaning of data presented graphically than data presented numerically in tables or frequency distributions. This is especially true if the users have little or no statistical knowledge.

Statistical graphs can be used to describe the data set or to analyze it. Graphs are also useful in getting the audience's attention in a publication or a speaking presentation. They can be used to discuss an issue, reinforce a critical point, or summarize a data set.



Histogram, frequency polygon, and cumulative frequency graph (ogive)

Procedure Table

Constructing a Histogram, Frequency Polygon, and Ogive

Step 1 Draw and label the x and y axes.

Step 2 On the x axis, label the class boundaries of the frequency distribution for the histogram and ogive. Label the midpoints for the frequency polygon.

Step 3 Plot the frequencies for each class, and draw the vertical bars for the histogram and the lines for the frequency polygon and ogive.

(*Note:* Remember that the lines for the frequency polygon begin and end on the x axis while the lines for the ogive begin on the x axis.)



Example 6: The histogram (revisit slide 50)

The **histogram** is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes.

Construct a histogram to represent the data shown for the record high temperatures for each of the 50 states.

Class boundaries	Frequency
99.5–104.5	2
104.5–109.5	8
109.5–114.5	18
114.5–119.5	13
119.5–124.5	7
124.5–129.5	1
129.5–134.5	1



Example 6

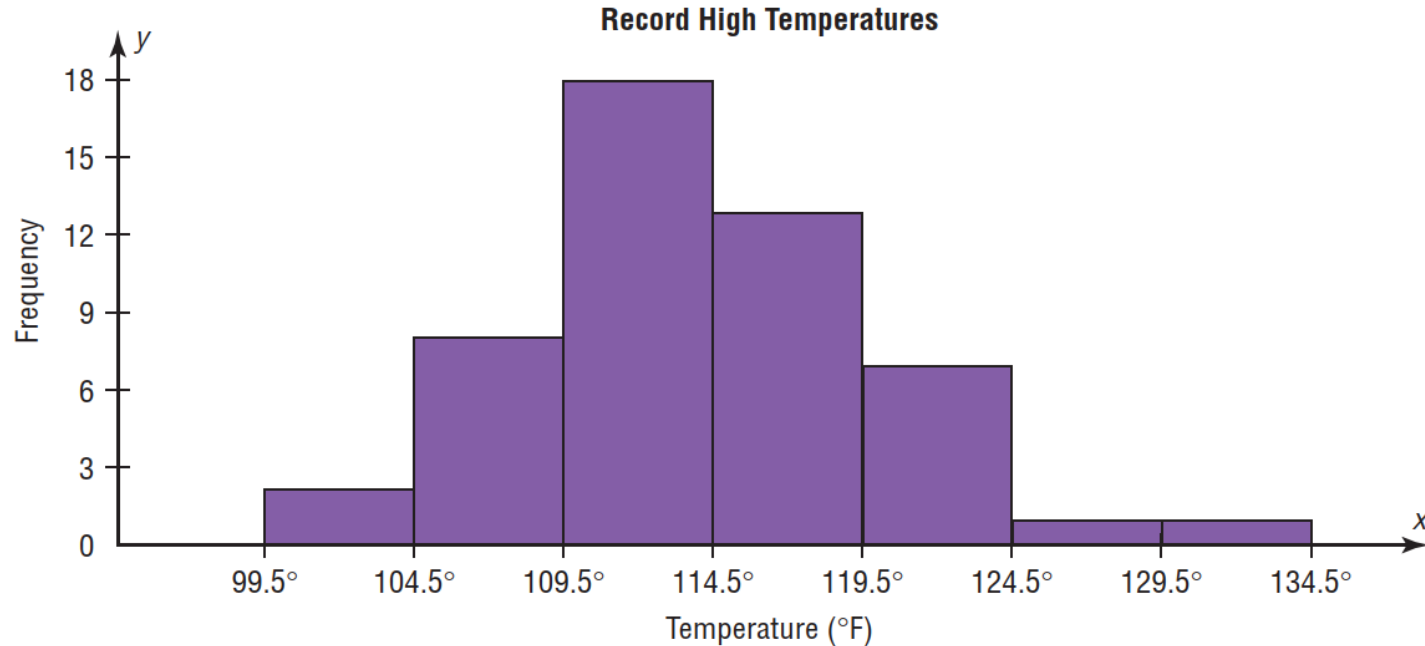
SOLUTION

- Step 1** Draw and label the x and y axes. The x axis is always the horizontal axis, and the y axis is always the vertical axis.
- Step 2** Represent the frequency on the y axis and the class boundaries on the x axis.
- Step 3** Using the frequencies as the heights, draw vertical bars for each class.



Example 6

SOLUTION



As the histogram shows, the class with the greatest number of data values (18) is 109.5–114.5, followed by 13 for 114.5–119.5. The graph also has one peak with the data clustering around it.

Example 7: The frequency polygon (revisit slide 50)

The **frequency polygon** is a graph that displays the data by using lines that connect points plotted for the frequencies at the midpoints of the classes. The frequencies are represented by the heights of the points.

Using the frequency distribution given, construct a frequency polygon.

Class boundaries	Frequency
99.5–104.5	2
104.5–109.5	8
109.5–114.5	18
114.5–119.5	13
119.5–124.5	7
124.5–129.5	1
129.5–134.5	1



Example 7

SOLUTION

Step 1 Find the midpoints of each class. Recall that midpoints are found by adding the upper and lower boundaries and dividing by 2:

$$\frac{99.5 + 104.5}{2} = 102 \qquad \frac{104.5 + 109.5}{2} = 107$$

and so on. The midpoints are

Class boundaries	Midpoints	Frequency
99.5–104.5	102	2
104.5–109.5	107	8
109.5–114.5	112	18
114.5–119.5	117	13
119.5–124.5	122	7
124.5–129.5	127	1
129.5–134.5	132	1

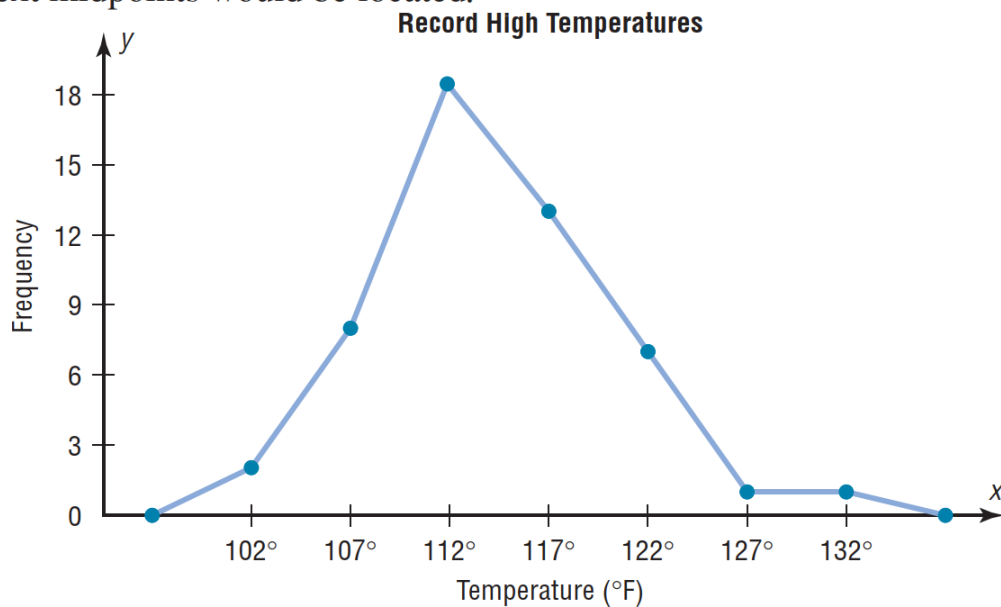
Step 2 Draw the x and y axes. Label the x axis with the midpoint of each class, and then use a suitable scale on the y axis for the frequencies.



Example 7

SOLUTION

- Step 3** Using the midpoints for the x values and the frequencies as the y values, plot the points.
- Step 4** Connect adjacent points with line segments. Draw a line back to the x axis at the beginning and end of the graph, at the same distance that the previous and next midpoints would be located.



The cumulative frequency graph (ogive)

The third type of graph that can be used represents the cumulative frequencies for the classes. This type of graph is called the *cumulative frequency graph*, or *ogive*. The **cumulative frequency** is the sum of the frequencies accumulated up to the upper boundary of a class in the distribution.

The **ogive** is a graph that represents the cumulative frequencies for the classes in a frequency distribution.



Example 8 (revisit slides 50, 57)

Using the frequency distribution given, construct an ogive.

Class boundaries	Frequency
99.5–104.5	2
104.5–109.5	8
109.5–114.5	18
114.5–119.5	13
119.5–124.5	7
124.5–129.5	1
129.5–134.5	1



Example 8

SOLUTION

Step 1 Find the cumulative frequency for each class.

	Cumulative frequency
Less than 99.5	0
Less than 104.5	2
Less than 109.5	10
Less than 114.5	28
Less than 119.5	41
Less than 124.5	48
Less than 129.5	49
Less than 134.5	50

Step 2 Draw the x and y axes. Label the x axis with the class boundaries. Use an appropriate scale for the y axis to represent the cumulative frequencies. (Depending on the numbers in the cumulative frequency columns, scales such as 0, 1, 2, 3, . . . , or 5, 10, 15, 20, . . . , or 1000, 2000, 3000, . . . can be used. Do *not* label the y axis with the numbers in the cumulative frequency column.) In this example, a scale of 0, 5, 10, 15, . . . will be used.

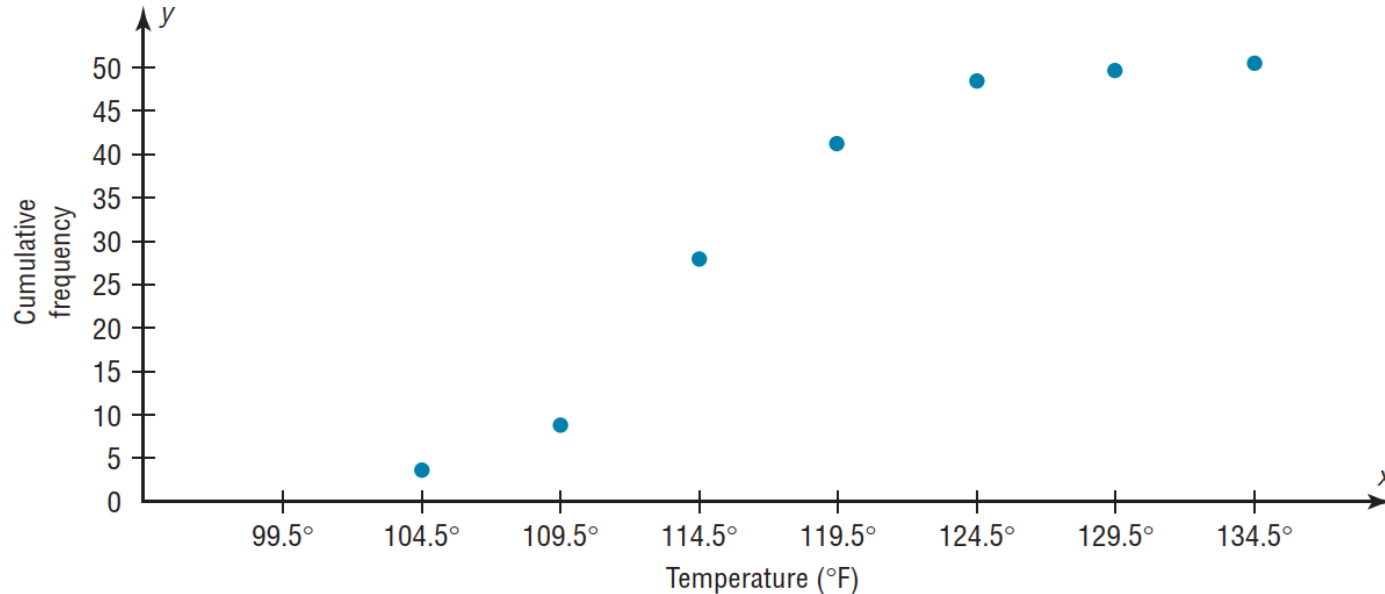


Example 8

SOLUTION

Step 3 Plot the cumulative frequency at each upper class boundary.

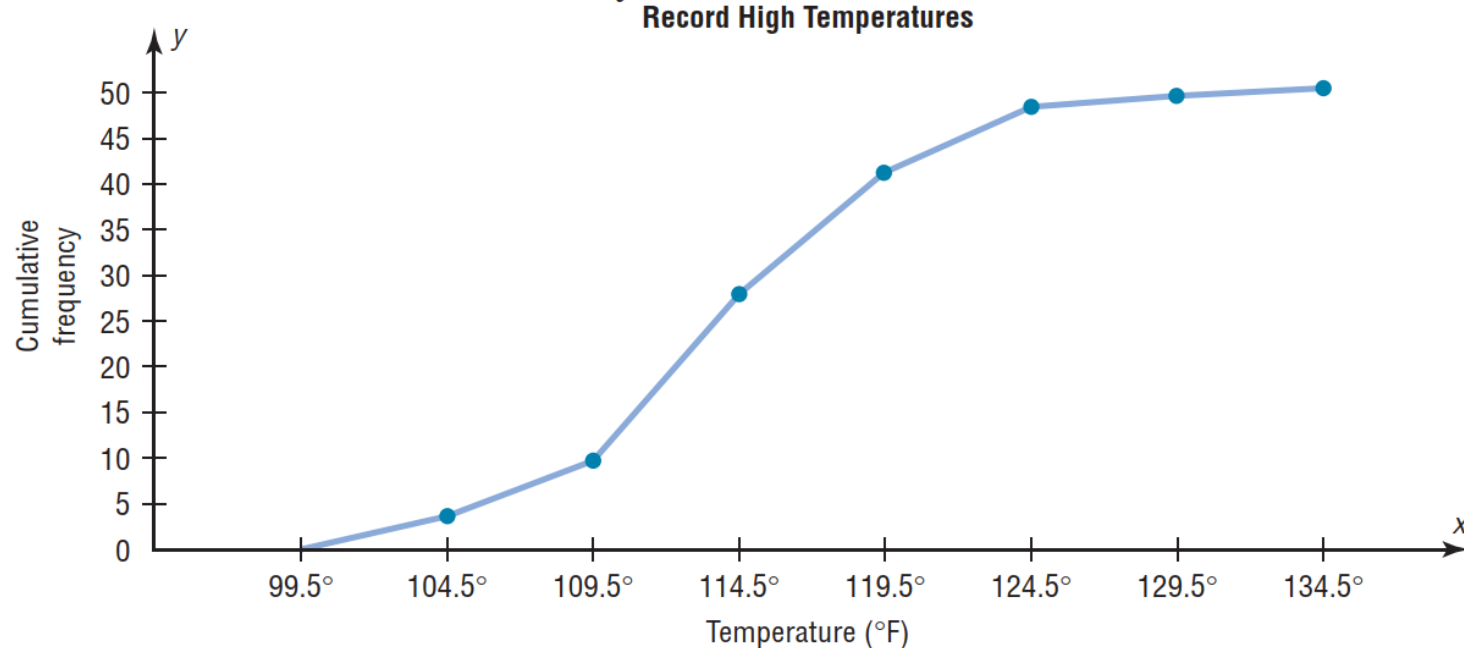
Upper boundaries are used since the cumulative frequencies represent the number of data values accumulated up to the upper boundary of each class.



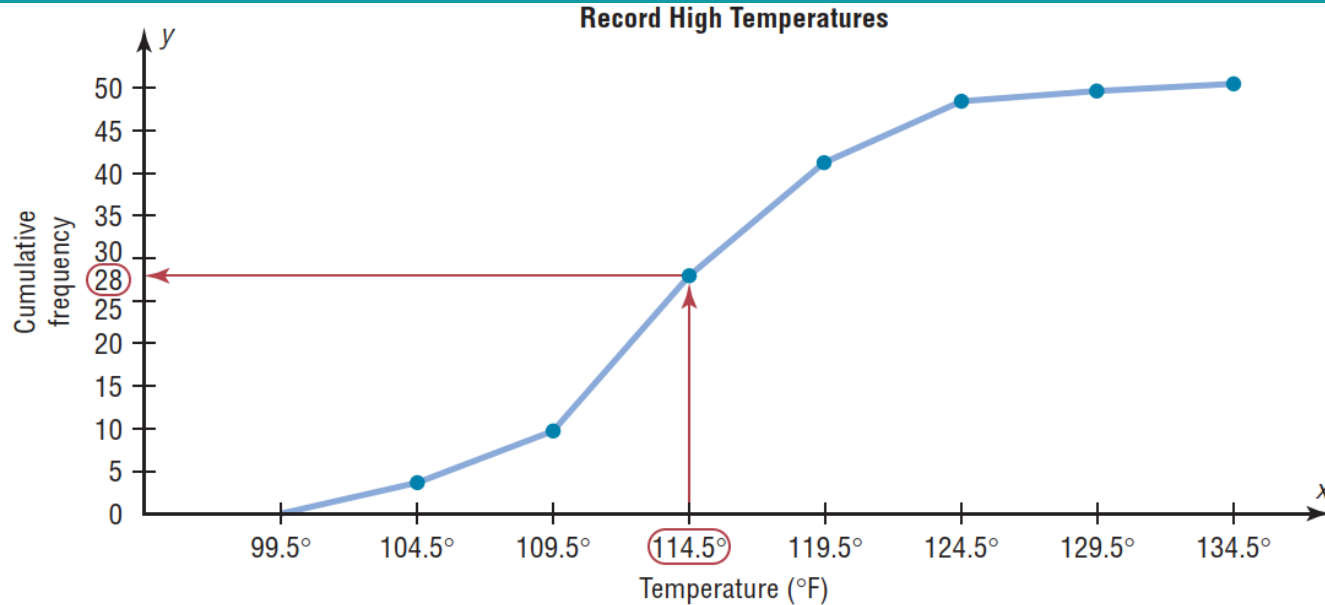
Example 8

SOLUTION

Step 4 Starting with the first upper class boundary, 104.5, connect adjacent points with line segments. Then extend the graph to the first lower class boundary, 99.5, on the x axis.



Finding a specific cumulative frequency



Cumulative frequency graphs are used to visually represent how many values are below a certain upper class boundary. For example, to find out how many record high temperatures are less than 114.5°F , locate 114.5°F on the x axis, draw a vertical line up until it intersects the graph, and then draw a horizontal line at that point to the y axis. The y axis value is 28.



Example 9

Construct a histogram, frequency polygon, and ogive using relative frequencies for the distribution (shown here) of the miles that 20 randomly selected runners ran during a given week.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	Total 20



Example 9

SOLUTION

Step 1 Convert each frequency to a proportion or relative frequency by dividing the frequency for each class by the total number of observations.

For class 5.5–10.5, the relative frequency is $\frac{1}{20} = 0.05$; for class 10.5–15.5, the relative frequency is $\frac{2}{20} = 0.10$; for class 15.5–20.5, the relative frequency is $\frac{3}{20} = 0.15$; and so on.

Place these values in the column labeled Relative frequency.

Class boundaries	Midpoints	Relative frequency
5.5–10.5	8	0.05
10.5–15.5	13	0.10
15.5–20.5	18	0.15
20.5–25.5	23	0.25
25.5–30.5	28	0.20
30.5–35.5	33	0.15
35.5–40.5	38	0.10
		Total 1.00



Example 9

SOLUTION

Step 2 Find the cumulative relative frequencies. To do this, add the frequency in each class to the total frequency of the preceding class. In this case, $0 + 0.05 = 0.05$, $0.05 + 0.10 = 0.15$, $0.15 + 0.15 = 0.30$, $0.30 + 0.25 = 0.55$, etc. Place these values in the column labeled Cumulative relative frequency.

An alternative method would be to find the cumulative frequencies and then convert each one to a relative frequency.

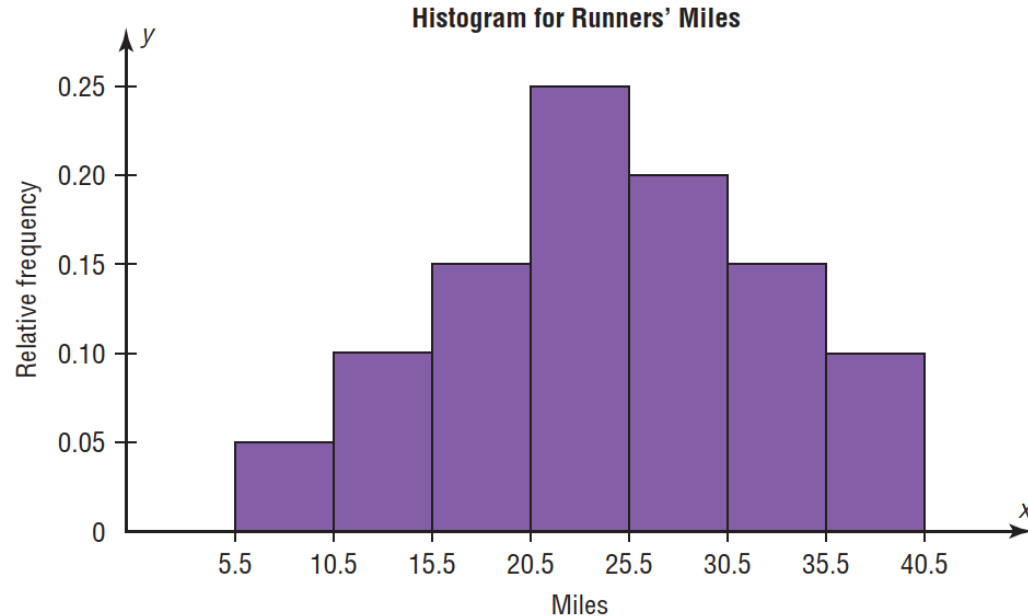
	Cumulative frequency	Cumulative relative frequency
Less than 5.5	0	0.00
Less than 10.5	1	0.05
Less than 15.5	3	0.15
Less than 20.5	6	0.30
Less than 25.5	11	0.55
Less than 30.5	15	0.75
Less than 35.5	18	0.90
Less than 40.5	20	1.00



Example 9

SOLUTION

Step 3 For the histogram and ogive, use the class boundaries along the x axis. For the frequency polygon, use the midpoints on the x axis. The scale on the y axis uses proportions.

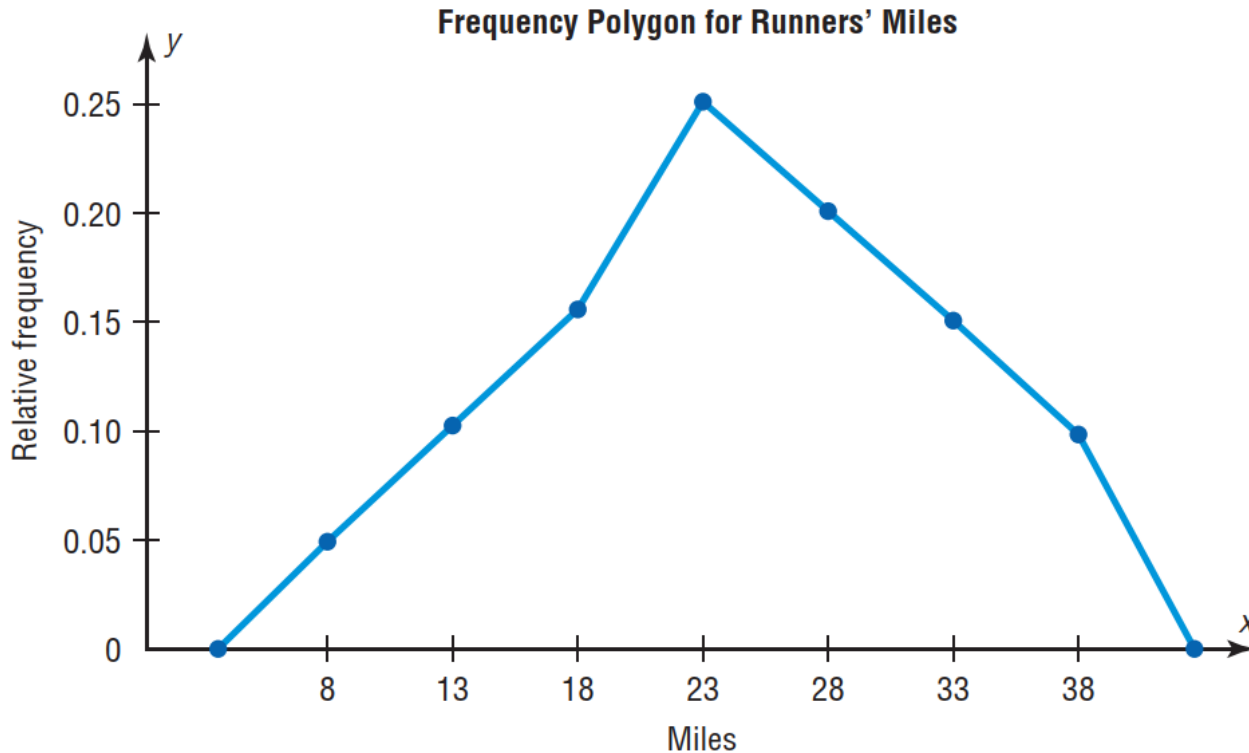


(a) Histogram



Example 9

SOLUTION

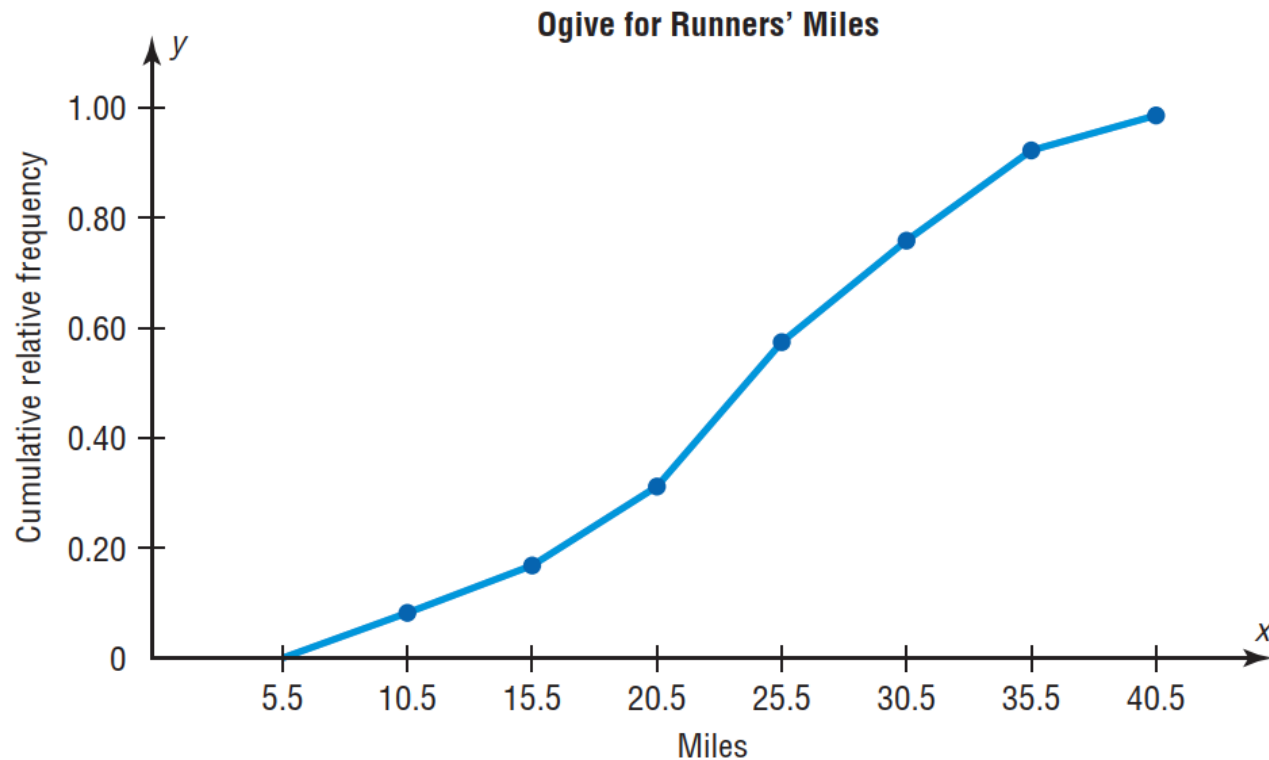


(b) Frequency polygon

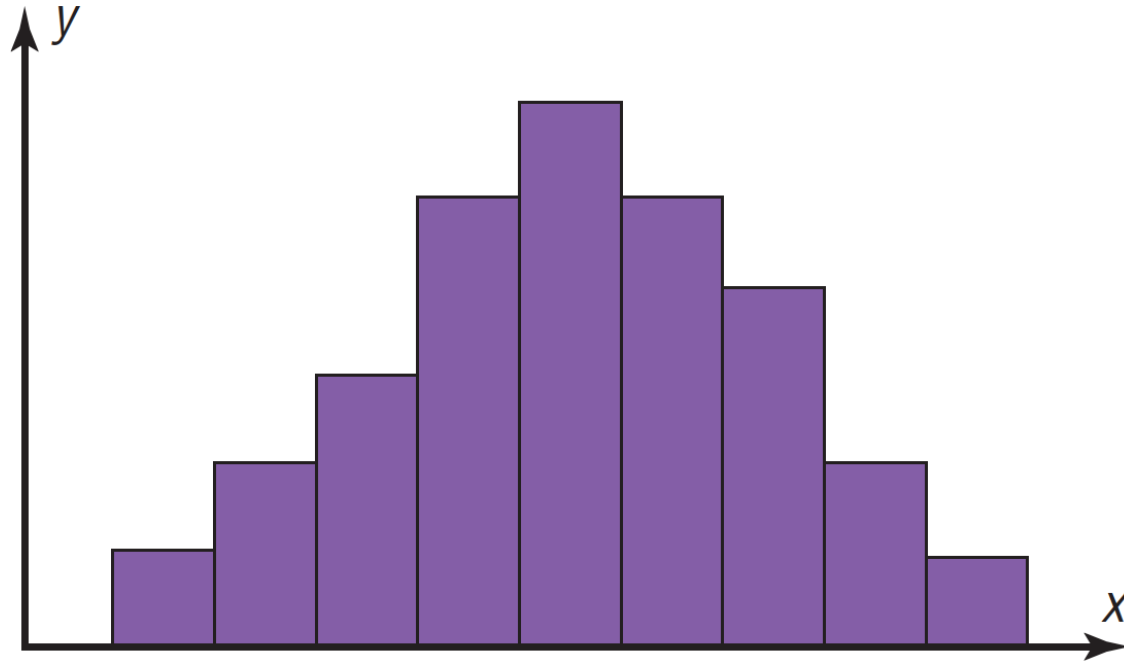


Example 9

SOLUTION



Distribution shapes

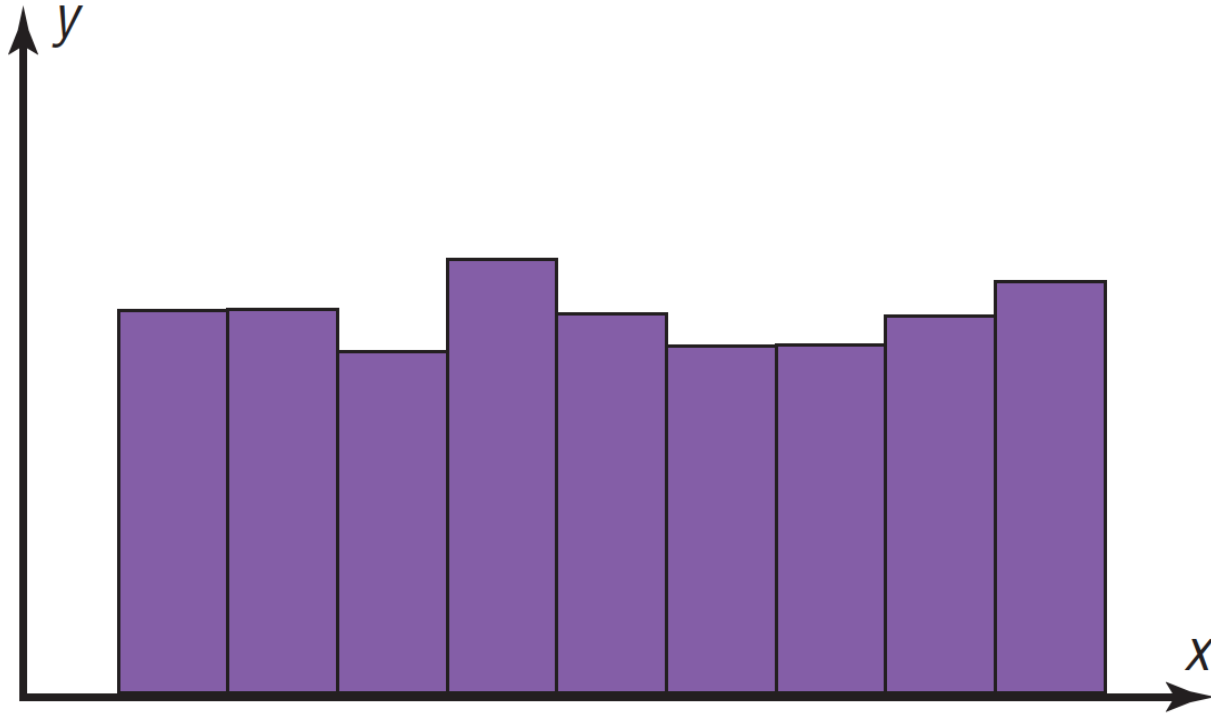


(a) Bell-shaped

A *bell-shaped distribution* shown has a single peak and tapers off at either end. It is approximately symmetric; i.e., it is roughly the same on both sides of a line running through the center.



Distribution shapes

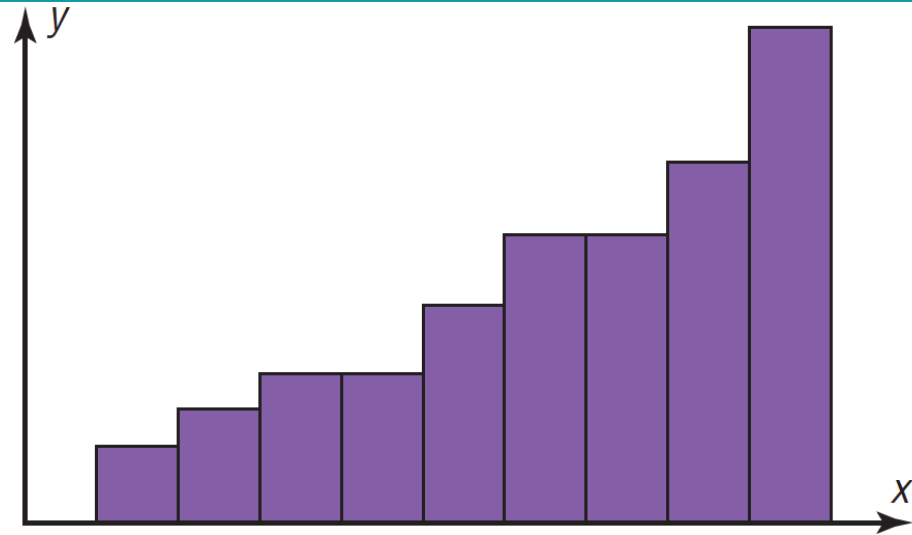


(b) Uniform

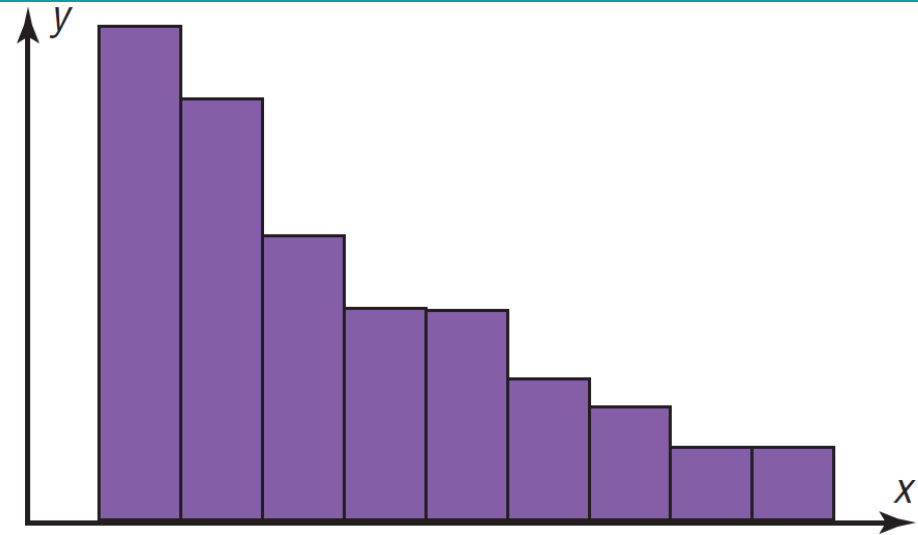
A uniform distribution is basically flat or rectangular.



Distribution shapes



(c) J-shaped

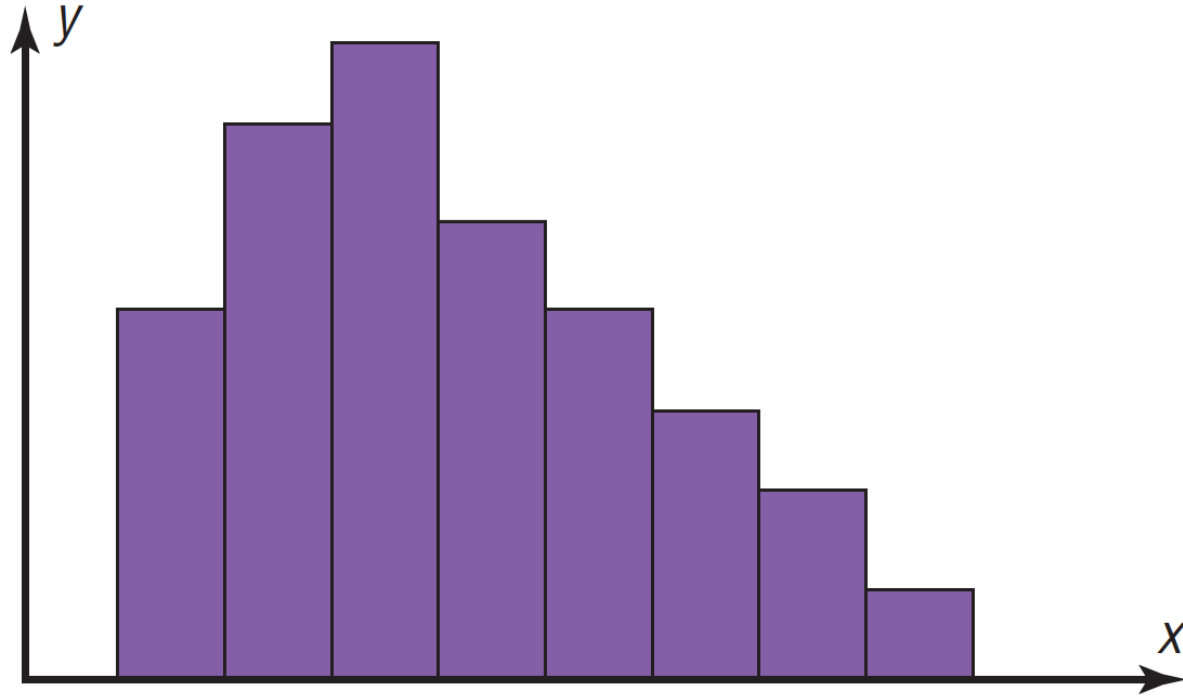


(d) Reverse J-shaped

A *J-shaped distribution* has a few data values on the left side and increases as one moves to the right. A *reverse J-shaped distribution* is the opposite of the J-shaped distribution.



Distribution shapes

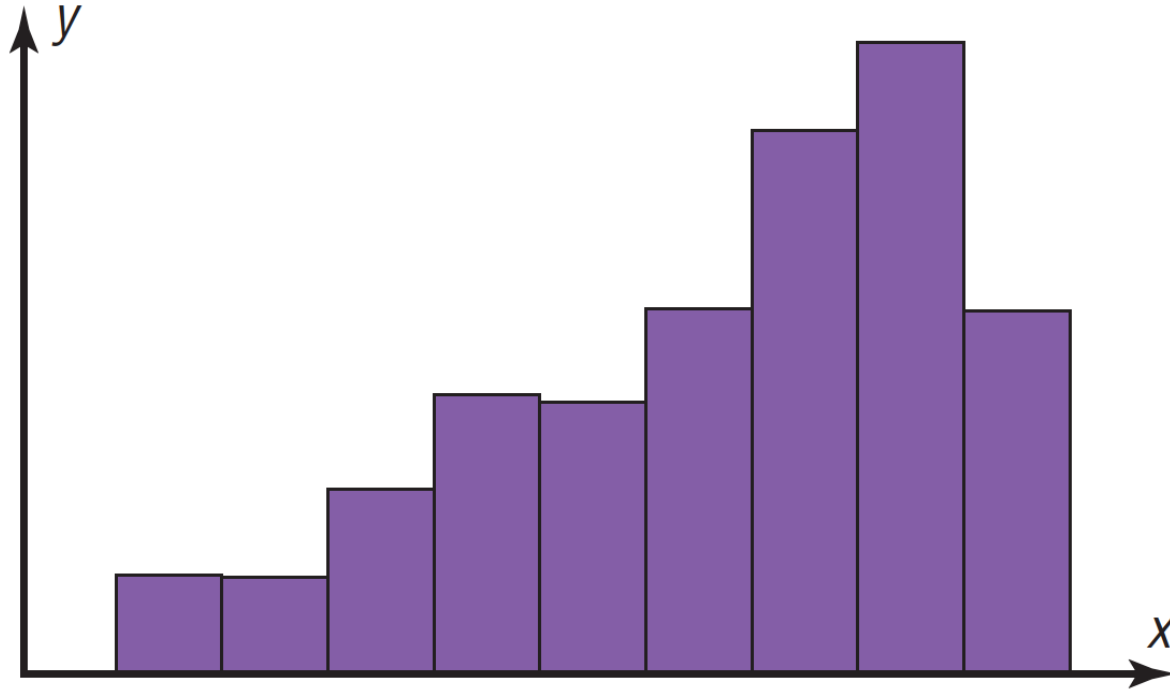


(e) Right-skewed

When the peak of a distribution is to the left and the data values taper off to the right, a distribution is said to be *positively or right-skewed*.



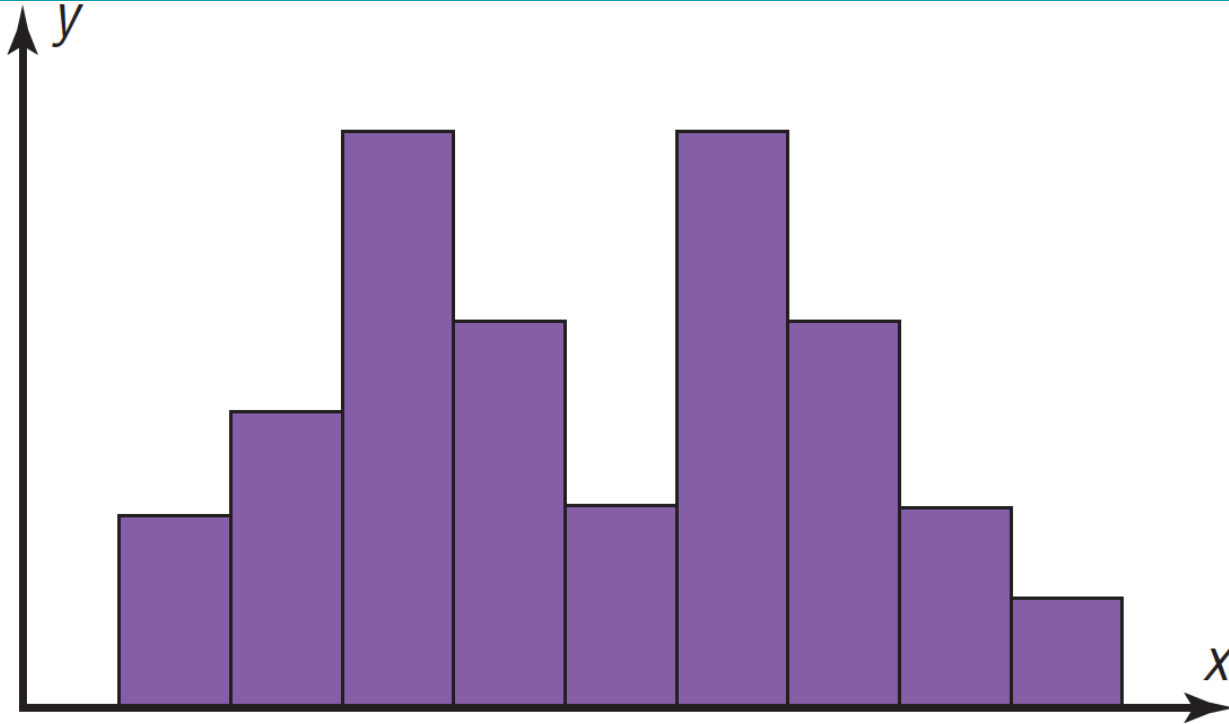
Distribution shapes



(f) Left-skewed



Distribution shapes

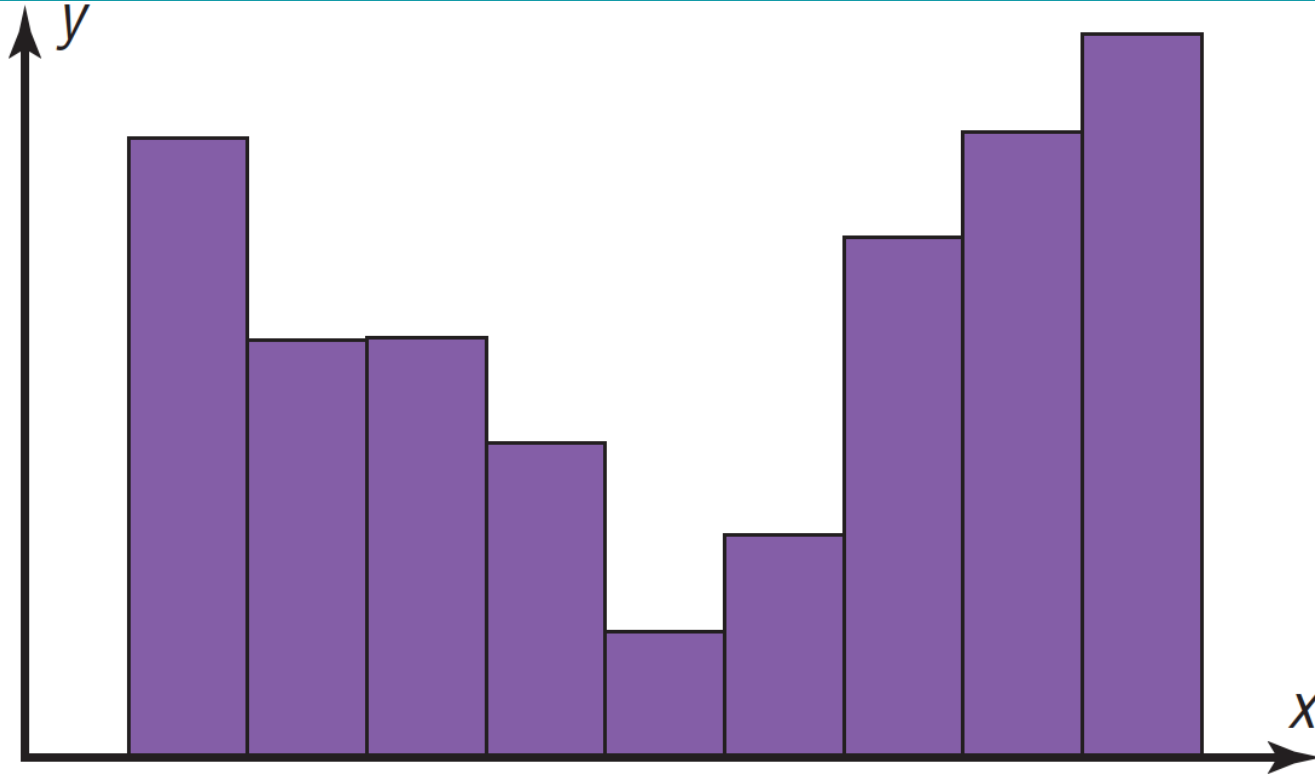


(g) Bimodal

When a distribution has two peaks of the same height, it is said to be *bimodal*.



Distribution shapes



(h) U-shaped

